

Grupa: 402
Specializarea: AIA
Nr.: 12
Data: 18.01.2021

Examen logică computațională

#1.

(a)

$$111101,101_{(2)} = ?_{(10)}$$

$$111101,101_{(2)} = 1 \cdot 2^{-3} + 0 \cdot 2^{-2} + 1 \cdot 2^{-1} + 1 \cdot 2^0 + 0 \cdot 2^1 + 1 \cdot 2^2 + 1 \cdot 2^3 + 1 \cdot 2^4 + 1 \cdot 2^5$$

$$= \frac{1}{8} + 0 + \frac{1}{2} + 1 + 0 + 4 + 8 + 16 + 32$$

$$= 0,125 + 0,5 + 61$$

$$= 61,625_{(10)}$$

$$61,625_{(10)} = ?_{(2)}$$

$$61 : 2 = 30 \text{ r } 1$$

$$30 : 2 = 15 \text{ r } 0$$

$$15 : 2 = 7 \text{ r } 1$$

$$7 : 2 = 3 \text{ r } 1$$

$$3 : 2 = 1 \text{ r } 1$$

$$1 : 2 = 0 \text{ r } 1$$

$$\Rightarrow 61_{(10)} = 111101_{(2)}$$

$$0,625 \cdot 2 = 0,25 + 1 \Rightarrow a_{-1} = 1$$

$$0,25 \cdot 2 = 0,5 + 0 \Rightarrow a_{-2} = 0$$

$$0,5 \cdot 2 = 0 + 1 \Rightarrow a_{-3} = 1$$

$$\Rightarrow 0,625_{(10)} = 0,101_{(2)}$$

$$\Rightarrow 61,625_{(10)} = 111101,101_{(2)}$$

$$\begin{aligned}
 15:2 &= 15 \text{ r } 1 \\
 14:2 &= 7 \text{ r } 0 \\
 7:2 &= 3 \text{ r } 1 \\
 3:2 &= 1 \text{ r } 1 \\
 1:2 &= 0 \text{ r } 1
 \end{aligned}$$

$$\Rightarrow 31_{(10)} = 11111_{(2)}$$

- prin mărime și semn (cod direct)

$$+31_{(10)} = 0001\ 1111_{(2)}$$

$$-31_{(10)} = 1001\ 1111_{(2)}$$

- prin complement față de 2 (cod complementar)

$$+31_{(10)} = 0001\ 1111_{(2)}$$

$$\begin{array}{r}
 \Downarrow \\
 1110\ 0000 \\
 \quad \quad \quad 1+ \\
 \hline
 1110\ 0001_{(2)} = -31_{(10)}
 \end{array}$$

- prin complement față de 1 (cod invers)

$$+31_{(10)} = 0001\ 1111_{(2)}$$

$$-31_{(10)} = 1110\ 0000_{(2)}$$

$$Y_1 = (X_1 + X_3)(X_2 + \bar{X}_4) + \bar{X}_2 \bar{X}_4$$

X_1	X_2	X_3	X_4	$\overline{X_1 X_2 X_3}$	$X_2 + \bar{X}_4$	$\overline{X_1 X_2 X_3} (X_2 + \bar{X}_4)$
1	1	1	1	1	1	1
1	1	1	0	1	1	1
1	1	0	1	0	1	0
1	1	0	0	0	1	0
1	0	1	1	1	0	0
1	0	1	0	1	1	1
1	0	0	1	1	0	0
1	0	0	0	1	1	1
0	1	1	1	1	1	1
0	1	1	0	1	1	1
0	1	0	1	1	1	1
0	1	0	0	1	1	1
0	0	1	1	1	0	0
0	0	1	0	1	1	1
0	0	0	1	1	0	0
0	0	0	0	1	1	1

X_1	X_2	X_3	X_4	$\bar{Y}_1 + X_3$	$X_2 + \bar{X}_4$	$(\bar{Y}_1 + X_3)(X_2 + \bar{X}_4)$	$\bar{Y}_2 \bar{X}_4$	$(\bar{Y}_1 + X_3)(X_2 + \bar{X}_4) + \bar{Y}_2 \bar{X}_4$
1	1	1	1	1	1	1	0	1
1	1	1	0	1	1	1	0	1
1	1	0	1	0	1	0	0	0
1	1	0	0	0	1	0	0	0
1	0	1	1	1	0	0	0	0
1	0	1	0	1	1	1	1	1
1	0	0	1	0	0	0	0	0
1	0	0	0	0	1	0	1	1
0	1	1	1	1	1	1	0	1
0	1	1	0	1	1	1	0	1
0	1	0	1	1	1	1	0	1
0	1	0	0	1	1	1	0	1
0	0	1	1	1	0	0	0	0
0	0	1	0	1	1	1	1	1
0	0	0	1	1	0	0	0	0
0	0	0	0	1	1	1	1	1

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3.

$$\overline{X_1 \bar{X}_2 X_4} \rightarrow \overline{X_2 X_3 \bar{X}_4} =$$

$$= \overline{X_1 \bar{X}_2 X_4} \cdot \overline{X_2 X_3 \bar{X}_4}$$

$$= (\bar{X}_1 + X_2 + \bar{X}_4) X_2 X_3 \bar{X}_4$$

$$= \bar{X}_1 X_2 X_3 \bar{X}_4 + X_2 X_3 \bar{X}_4 + X_2 X_3 \bar{X}_4$$

$$4. f(X_1, X_2, X_3, X_4) = X_1 \bar{X}_3 + X_2 + X_1 X_2 \bar{X}_3$$

FCD

$$f(X_1, X_2, X_3, X_4) = X_1 \bar{X}_3 (X_2 + \bar{X}_2) + X_2 (X_1 + \bar{X}_1) + X_1 X_2 \bar{X}_3$$

FCD

$$f(X_1, X_2, X_3, X_4) = X_1 X_2 \bar{X}_3 + X_1 \bar{X}_2 \bar{X}_3 + X_1 X_2 + \bar{X}_1 X_2 + X_1 X_2 \bar{X}_3$$

FCD

$$f(X_1, X_2, X_3, X_4) = X_1 X_2 \bar{X}_3 + X_1 \bar{X}_2 \bar{X}_3 + X_1 X_2 (X_3 + \bar{X}_3) + \bar{X}_1 X_2 (X_3 + \bar{X}_3)$$

FCD

$$f(X_1, X_2, X_3, X_4) = X_1 X_2 \bar{X}_3 + X_1 \bar{X}_2 \bar{X}_3 + X_1 X_2 X_3 + X_1 X_2 \bar{X}_3 + \bar{X}_1 X_2 X_3 + \bar{X}_1 X_2 \bar{X}_3$$

FCD

$$f(X_1, X_2, X_3, X_4) = X_1 X_2 \bar{X}_3 + X_1 \bar{X}_2 \bar{X}_3 + X_1 X_2 X_3 + \bar{X}_1 X_2 X_3 + \bar{X}_1 X_2 \bar{X}_3$$

5.

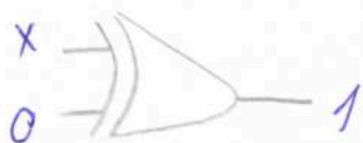
(a)

$$\bar{x} \cdot 0 = 1 \Rightarrow \overline{1 \cdot \bar{x} \cdot 0} = 1 \Rightarrow \bar{1} + \bar{\bar{x}} + \bar{0} = 1 \Rightarrow 0 + x + 1 = 1 \Rightarrow$$

(1)

$\Rightarrow$ Valorile pe care le poate lua variabila x sunt: 0 sau 1

a	b	c	f
1	1	1	0
1	1	0	1
1	0	1	1
1	0	0	1
0	1	1	1
0	1	0	1
0	0	1	1
0	0	0	1



(2)

$$\Rightarrow x \oplus 0 = 1 \Rightarrow \bar{x} \cdot 0 + x \cdot \bar{0} = 1 \Rightarrow$$

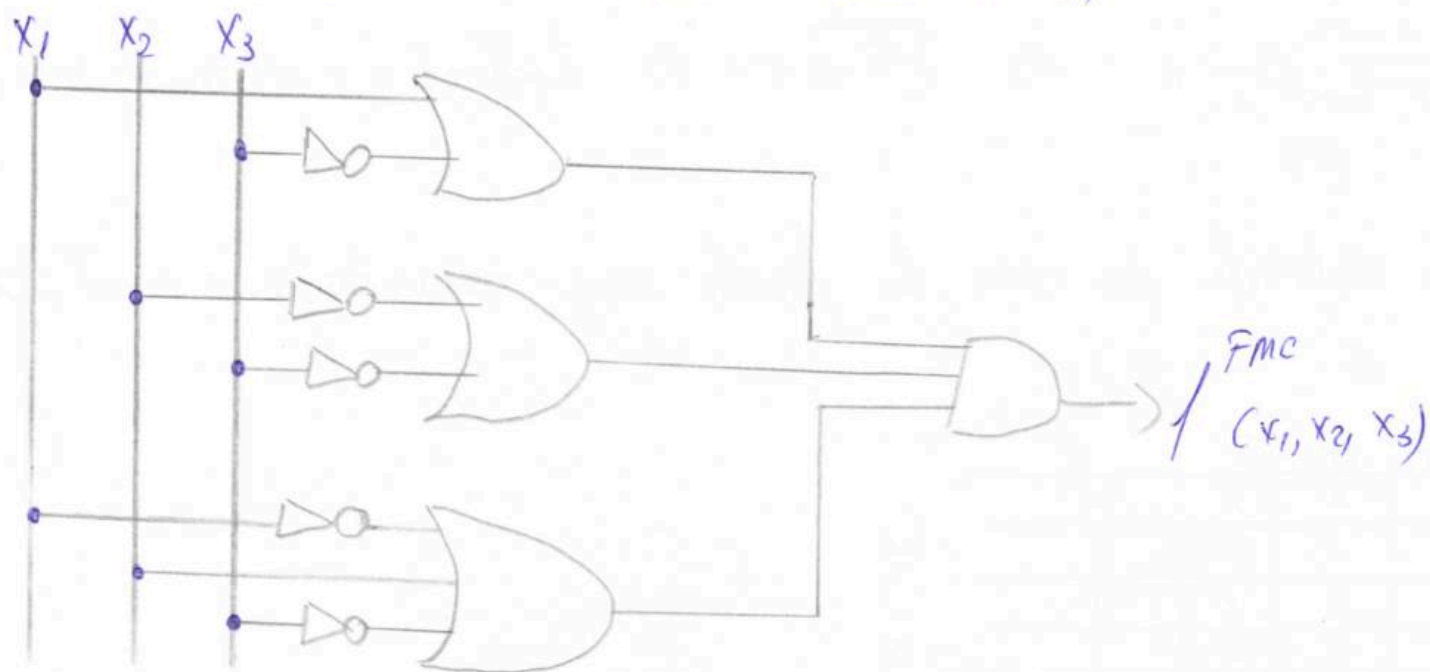
$$\Rightarrow 0 + x \cdot 1 = 1 \Rightarrow x = 1 \Rightarrow$$

$\Rightarrow$ Valorile pe care le poate lua x sunt: 1

a	b	f
1	1	0
1	0	1
0	1	1
0	0	0

(b) $\swarrow$ FMC

$$f(x_1, x_2, x_3) = (x_1 + \bar{x}_3)(\bar{x}_2 + \bar{x}_3)(\bar{x}_1 + \dots)$$



6.

(a) $\swarrow$ FCD

$$f(x_1, x_2, x_3, x_4) = \sum(1, 3, 5, 9, 12, 13, 14, 15)$$

$x_3 x_4 \backslash x_1 x_2$	00	01	11	10
00	0	0	1	0
01	1	1	1	1
11	1	0	1	0
10	0	0	1	0

$\swarrow$ FMD

$$f(x_1, x_2, x_3, x_4) = \bar{x}_3 x_4 + x_1 x_2 + \bar{x}_1 \bar{x}_2 x_4$$

$\swarrow$ FMD

$$f(x_1, x_2, x_3, x_4) = \bar{x}_1 \bar{x}_4 + \bar{x}_1 x_2 x_3 + \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 x_3 + x_1 \bar{x}_2 \bar{x}_4$$

$\swarrow$ FMC

$$f(x_1, x_2, x_3, x_4) = (x_1 + x_4)(x_1 + \bar{x}_2 + \bar{x}_3)(\bar{x}_1 + x_2 + \bar{x}_3)(\bar{x}_1 + x_2 + x_4)$$

	grupă	ter. cuplate	echivalentul binar				Impl. primi
			x_1	x_2	x_3	x_4	
3 = 0011	0	1	0	0	0	1	✓
5 = 0101		3	0	0	1	1	✓
9 = 1001		5	0	1	0	1	✓
12 = 1100		9	1	0	0	1	✓
13 = 1101		12	1	1	0	0	✓
14 = 1110	2	13	1	1	0	1	✓
15 = 1111		14	1	1	1	0	✓
	3	15	1	1	1	1	✓

Grupă	Indicii ter. cuplate	Echivalentul binar				Impl. primi
		x_1	x_2	x_3	x_4	
0	1, 3	0	0	-	1	α
	1, 5	0	0	0	1	✓
	1, 9	-	0	0	1	✓
1	5, 13	-	1	0	1	β
	9, 13	1	-	0	1	δ
	12, 13	1	1	0	-	✓
	12, 14	1	1	-	0	✓
2	13, 15	1	1	-	1	✓
	14, 15	1	1	1	-	✓

Grupă	Indicii ter. cuplate	Echivalentul binar				Impl. primi
		x_1	x_2	x_3	x_4	
0	1, 5, 9, 13	-	-	0	1	Δ
	1, 9, 5, 13	-	-	0	1	Δ
1	12, 13, 14, 15	1	1	-	-	γ
	12, 14, 13, 15	1	1	-	-	γ

Imp. combina- zioni	1	3	5	9	12	13	14	15
$\angle$ 1, 3	*	*						
β 5, 13			*			*		
γ 9, 13				*		*		
Δ 1, 5, 9, 13	*		*	*		*		
Υ 12, 13, 14, 15					*	*	*	*

iPF

iPF

FMD

$$(x_1, x_2, x_3, x_4) = \bar{x}_1 \bar{x}_2 x_4 + x_1 x_2 + \bar{x}_3 x_4$$

(b)
$$(x_1, x_2, x_3, x_4) = \sum (3, 5, 9, 12, 14, 15)$$

$$d = 1 = 6 = 13 = *$$

$x_3 x_4$	00	01	11	10
00	0	0	1	0
01	d	1	d	1
11	1	0	1	0
10	0	d	1	0

FMD

$$(x_1, x_2, x_3, x_4) = \bar{x}_3 x_4 + x_1 x_2 + \bar{x}_1 x_3 \bar{x}_4$$

FMD

$$(x_1, x_2, x_3, x_4) = \bar{x}_3 x_4 + x_1 x_2 + x_2 x_3 \bar{x}_4 + \bar{x}_1 \bar{x}_2 x_4$$

FMD

$$(x_1, x_2, x_3, x_4) = \bar{x}_1 \bar{x}_2 + \bar{x}_1 x_2 x_3 + \bar{x}_1 \bar{x}_2 \bar{x}_3 + \bar{x}_2 \bar{x}_3 \bar{x}_4 + x_1 \bar{x}_2 x_3$$

FMD

$$(x_1, x_2, x_3, x_4) = (x_1 + x_4)(x_1 + \bar{x}_2 + \bar{x}_3)(x_1 + x_2 + x_3)(x_2 + x_3 + x_4)$$

$$+ (\bar{x}_1 + x_2 + \bar{x}_3)$$